

	Work Instruction	Doc. Revision	03
	AXI V810 X-Axis Ball Screw Removal & Replacement Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	28 Mar 2012	First Release	ONG KEAN SHENG
01	09 Apr 2015	Second Release	Muhammed Shameem
02	2 May 2016	Content Update	KOBE TAN
03	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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X-Axis Ball Screw Removal

1. Remove the rotary encoder cable from the motor as shown in Figure 1.

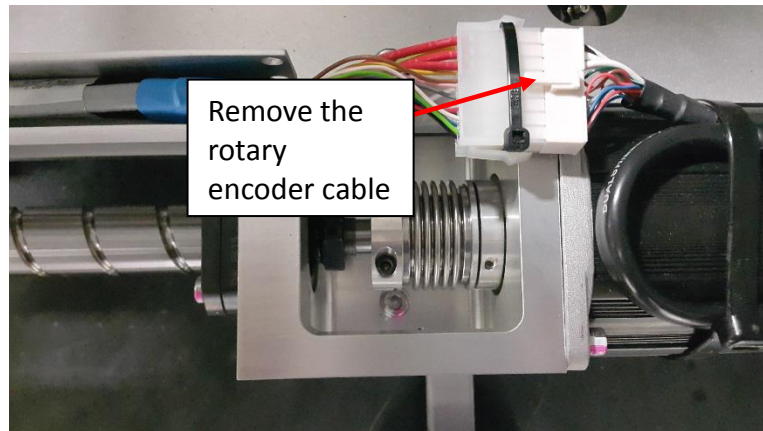


Figure 1 Rotary encoder

2. Loosen the 2 coupling screws at the coupling(2660-0032) as shown in Figure 2.

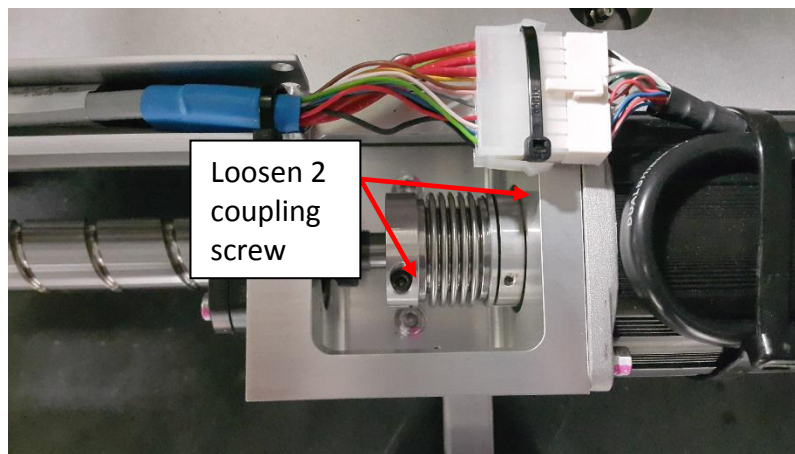


Figure 2 Coupling

3. Remove the 4 x M4 screw at the motor mount and remove the motor from the X-axis stage as shown in the Figure 3.

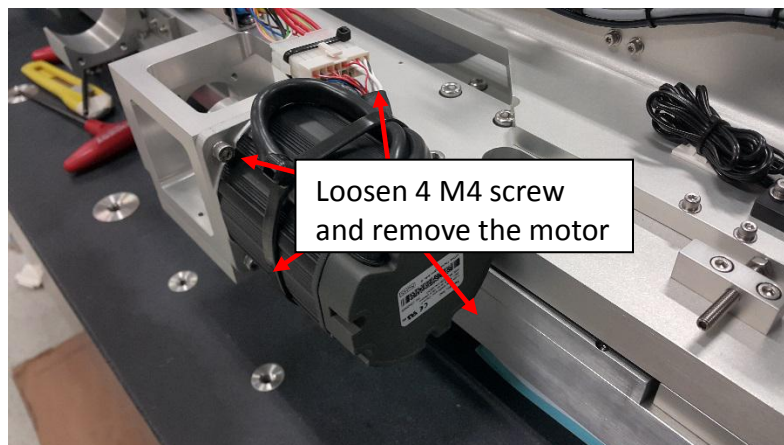


Figure 3 X-Axis Motor

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- Remove the coupling shown in Figure 4.

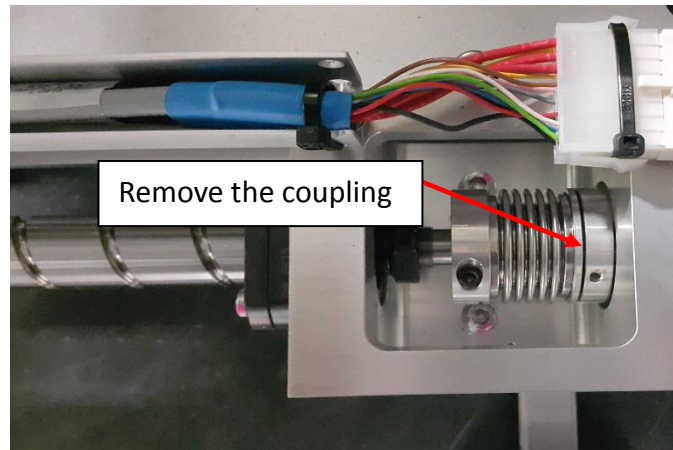


Figure 4 Motor Mount

- Remove the screw at "A" & "B" of the Bearing Support with M6 Allen Key as shown in Figure 5.
- Remove the screw at "G" to "J" of the Fixed Support Bearing with M5 Allen Key as shown in Figure 5.
- Remove the screw at "C" to "F" of the Fixed Support Bearing with M6 Allen Key as shown in Figure 5.

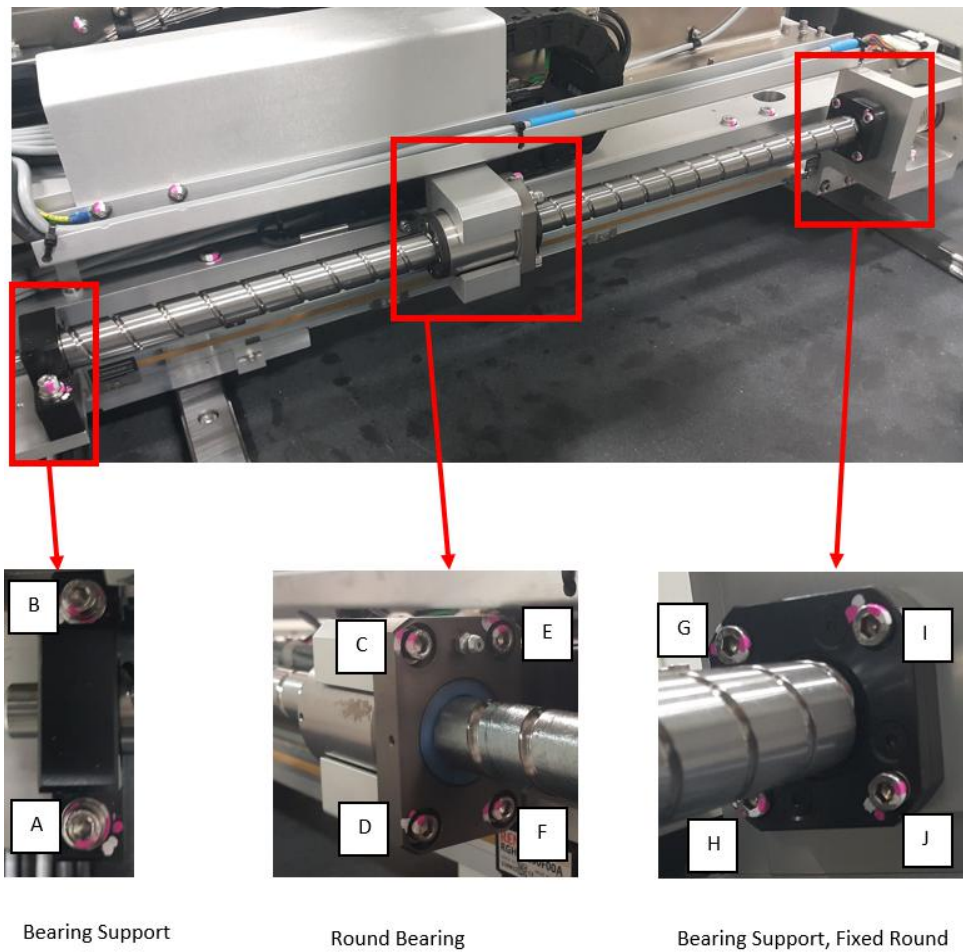


Figure 5 Ball screw

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8. Gently remove all screw from the X-Axis stage as shown in Figure 6.

Remarks : Caution on handling especially on critical component (Encoder & Glass Scale)

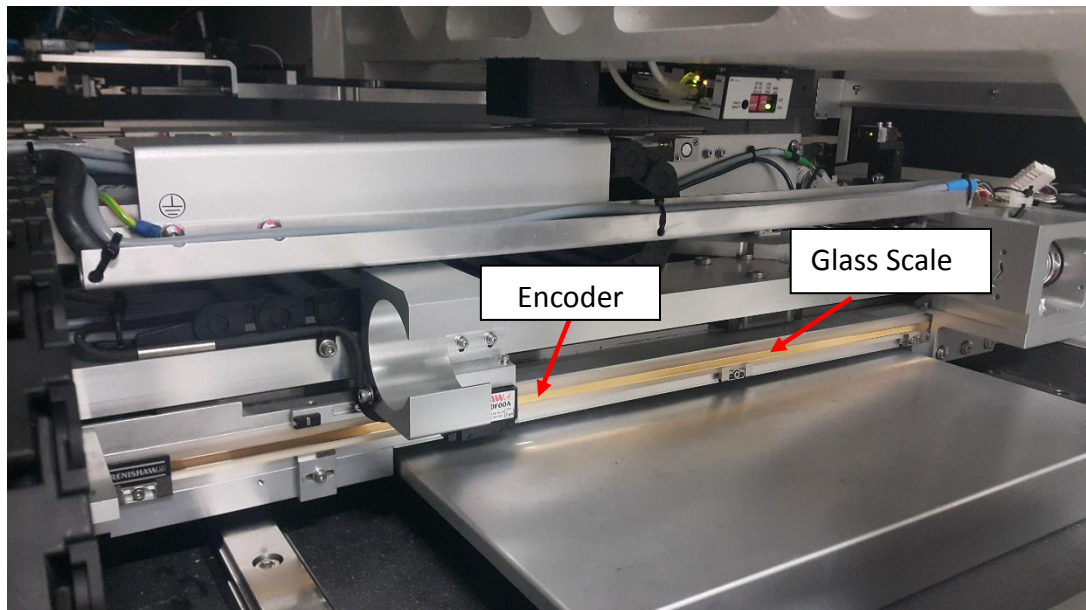


Figure 6 Read Head and Glass Scale

9. Place ball screw on the flat & Clean surface to ease of removal.



Figure 7 Ball Screw with bearings

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10. Using M2 Allen Key, unscrew the set screw as shown in Figure 8.

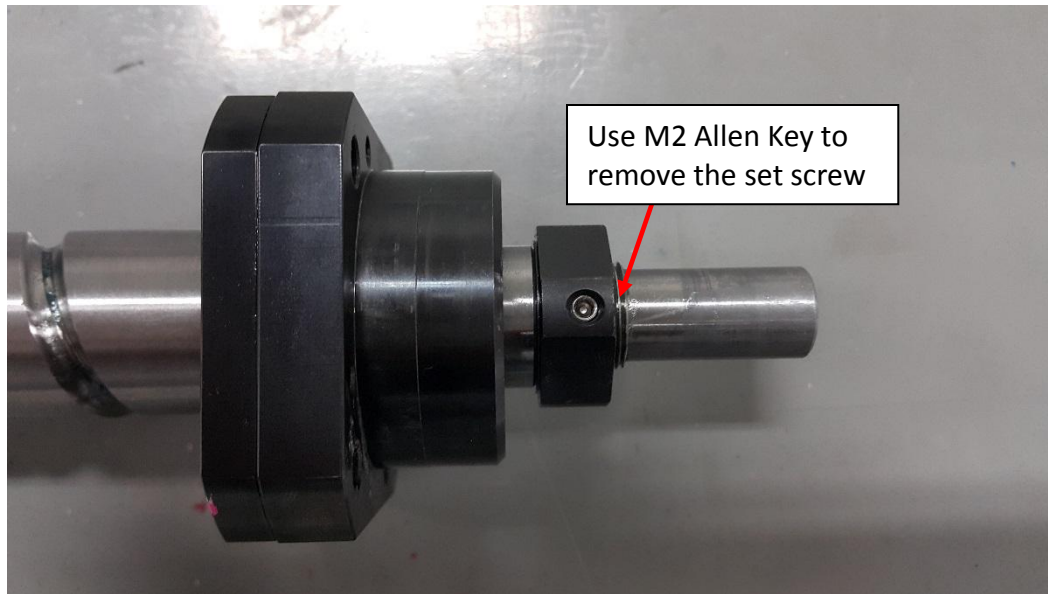


Figure 8 Set Screw

11. Gently remove Gold Nut as below Figure.

Remarks : Caution on handling especially on Gold Nut.

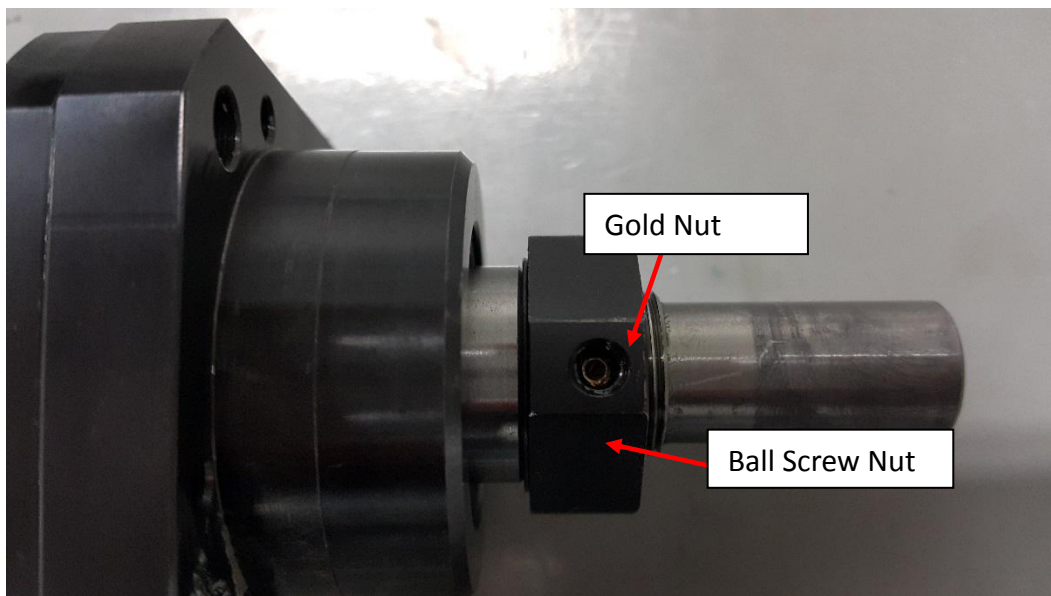


Figure 9 Ball Screw Gold Nut

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12. Gently insert 4XACM-A102, Ball Screw Tightening Jig (Figure 10) at the end of the ball screw as shown in Figure 11.

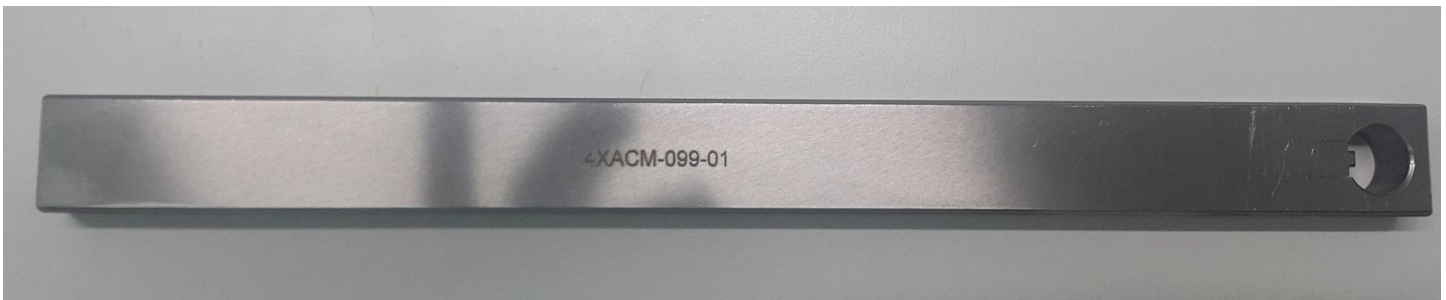


Figure 10 4XACM-A102 Ball Screw Tightening Jig

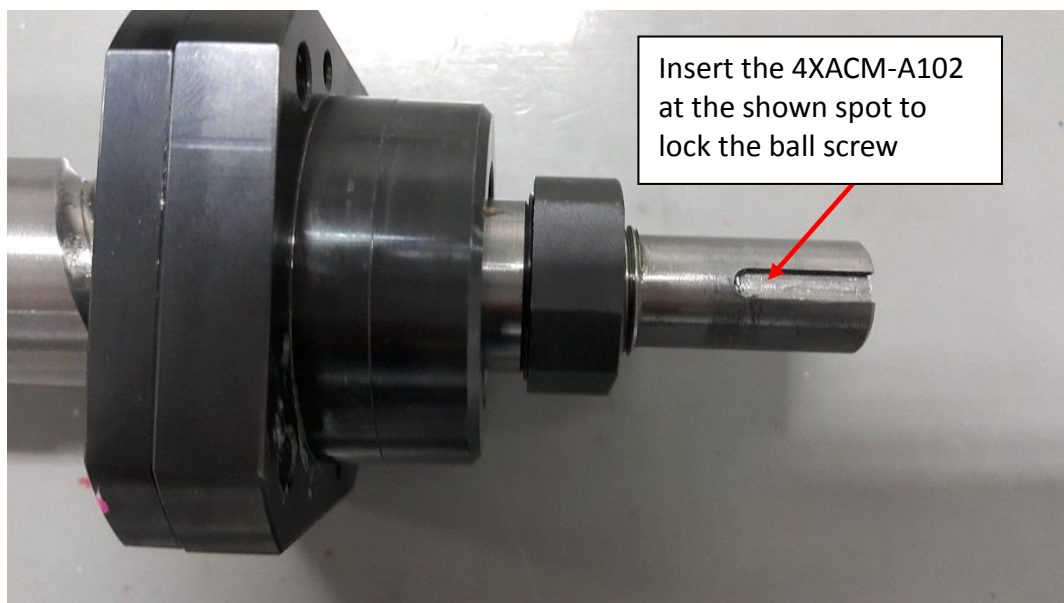


Figure 11 Ball screw tightening jig locking position

13. Use Size 22 spanner to hold ball screw nut as shown in Figure 12.

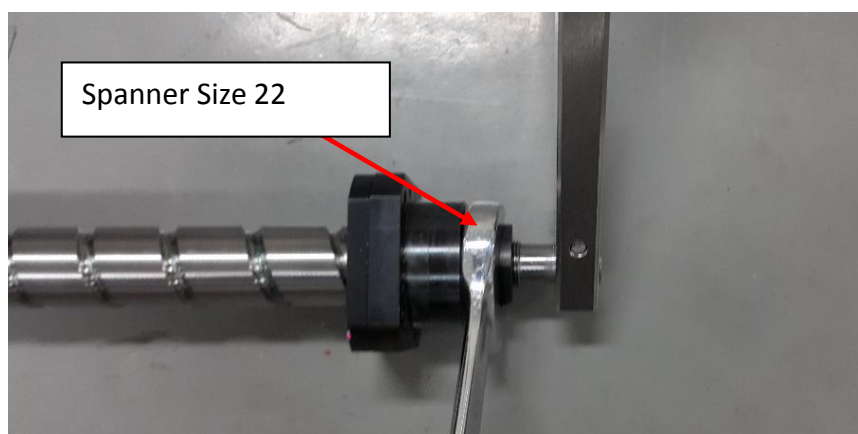


Figure 12 Ball Screw Nut

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14. Hold the 4XACM-A102 at the fixed position; rotate the spanner in the direction of anti-clockwise to remove the nut as shown in Figure 13.

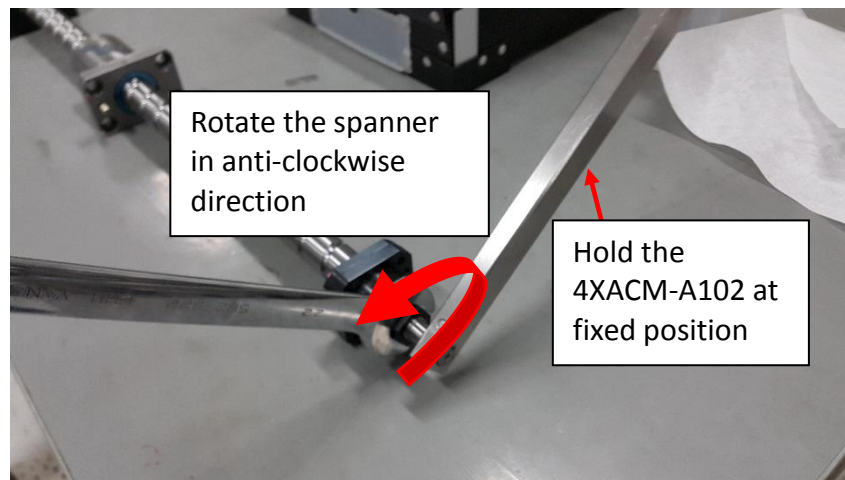


Figure 13 Ball Screw Nut Opening

15. Gently position End Bearing Support Nut , Gold Nut & M2 Set Screw at visible area .May refer to Figure 14.

Remarks : Caution on handling as below parts would require for re-use.

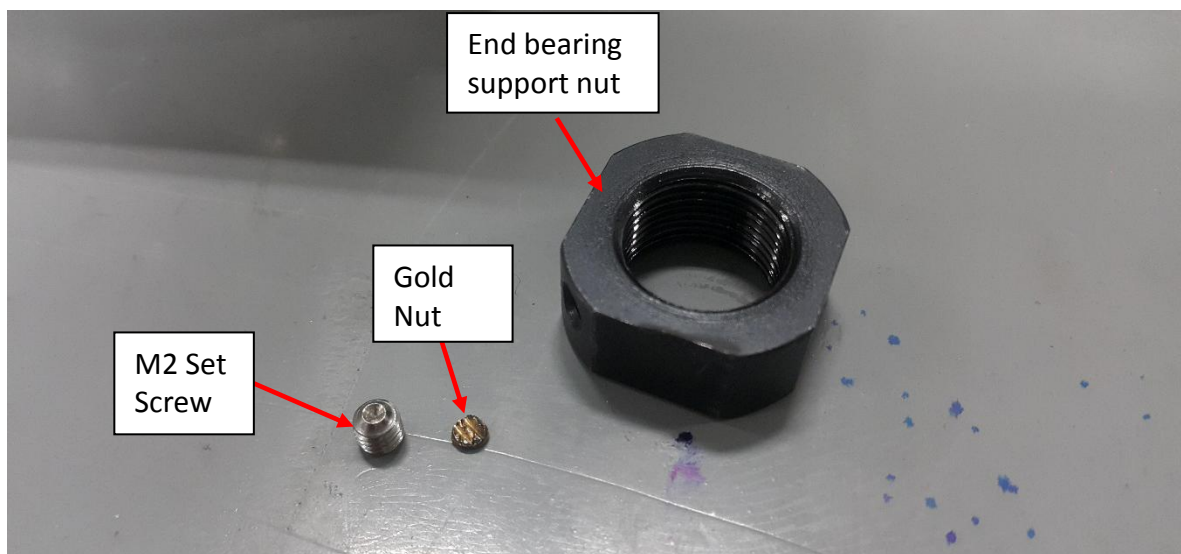


Figure 14 Gold Nut, M2 Set Screw & End Bearing Support Nut

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16. Gently remove silver spacer as below Figure 15.

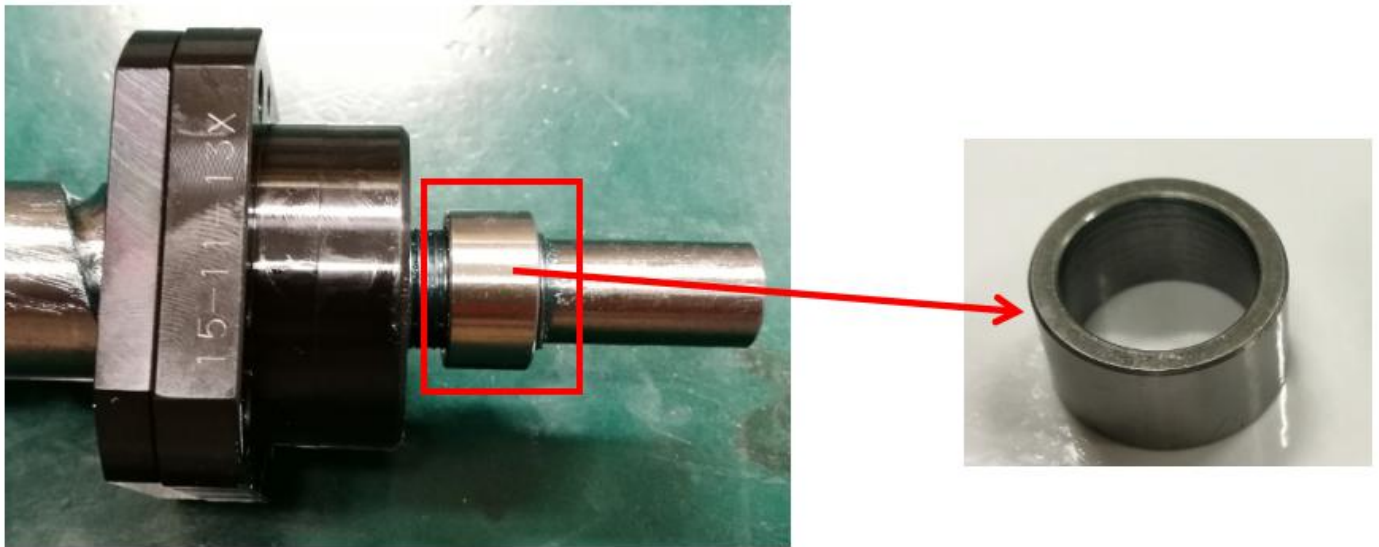


Figure 15 Silver Spacer Removal

17. Gently remove the fixed end support bearing from the ball screw as shown in Figure 16.

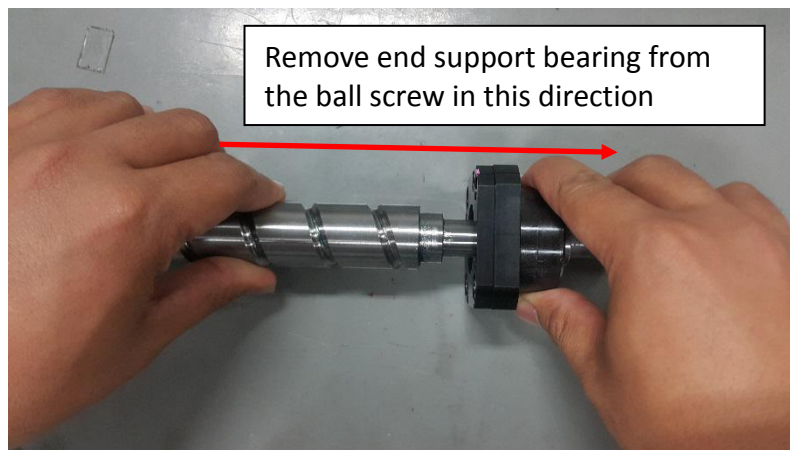


Figure 16 End support bearing removal



Figure 17 End support bearing removal [COMPLETE]

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18. Gently remove other end of bearing block from ball screw. May refer to below Figure 18.

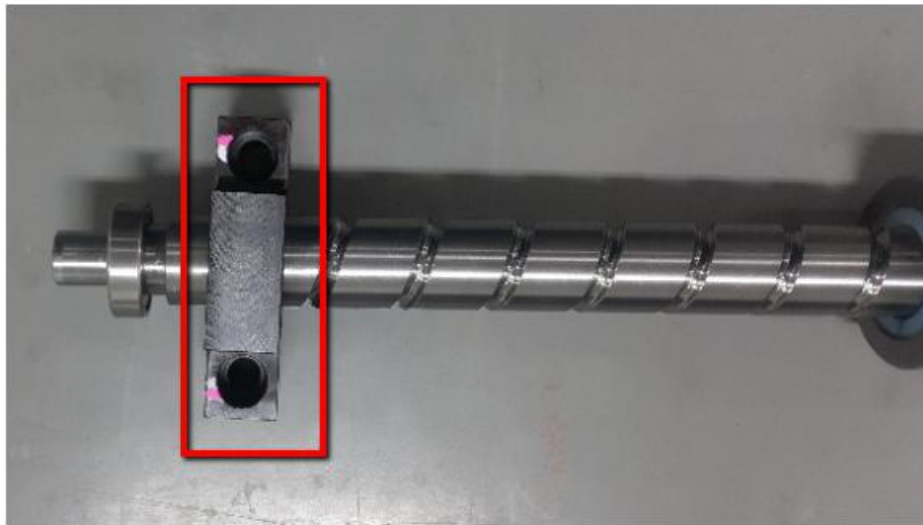


Figure 18 End bearing block removal

19. Prepare C-Clip Plier (Figure 19) to remove the C-Clip from the ball screw end.



Figure 19 C-Clip Plier

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20. Remove the C-Clip from the ball screw as shown in Figure 20 to Figure 23.

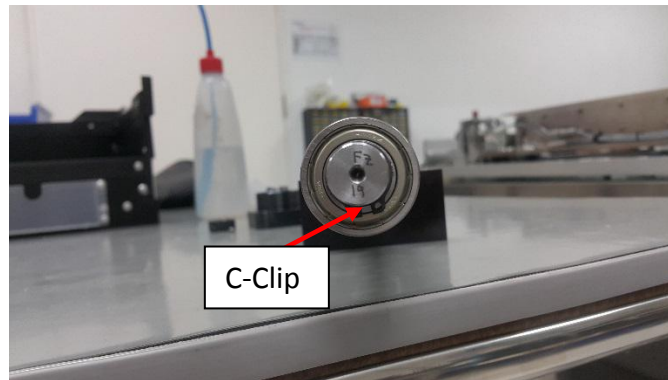


Figure 20 C-Clip

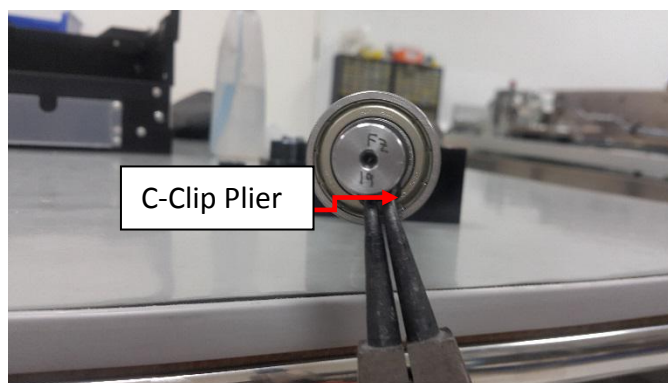


Figure 21 C-Clip Plier

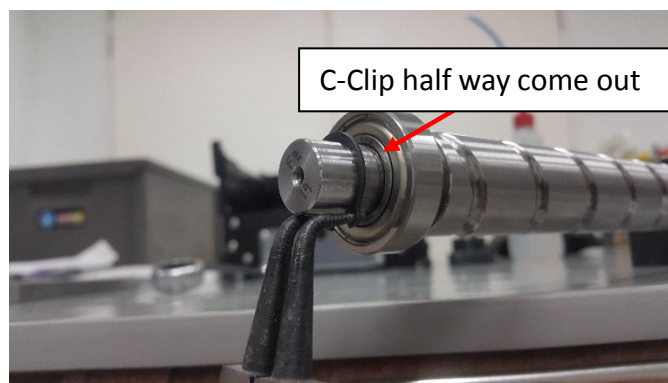


Figure 22 C-Clip Plier (2)

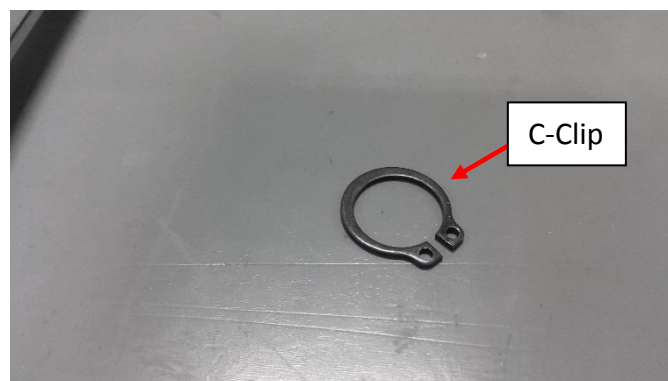


Figure 23 C-Clip (Removal Complete)

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21. By using Plier rubber holder and gently remove the bearing as shown in Figure 24 to Figure 27.

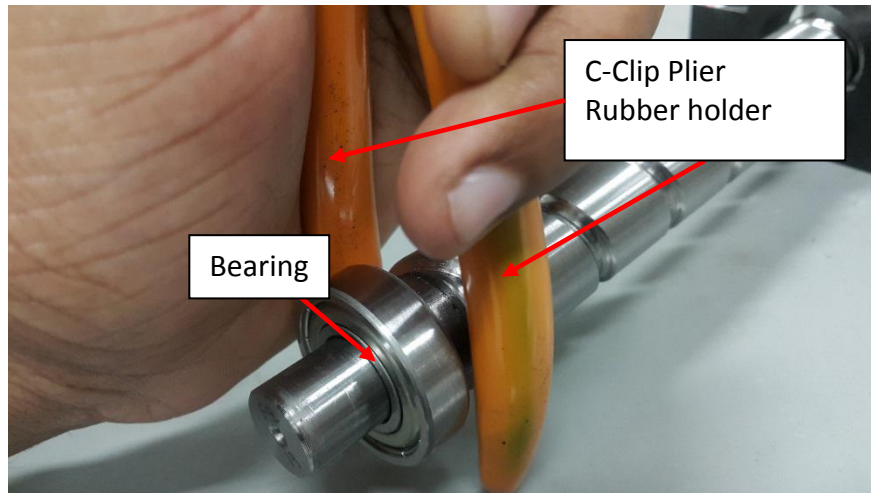


Figure 24 End Bearing

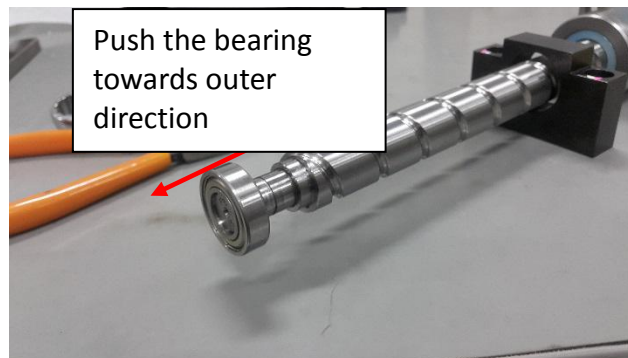


Figure 25 Bearing at tip of end shaft

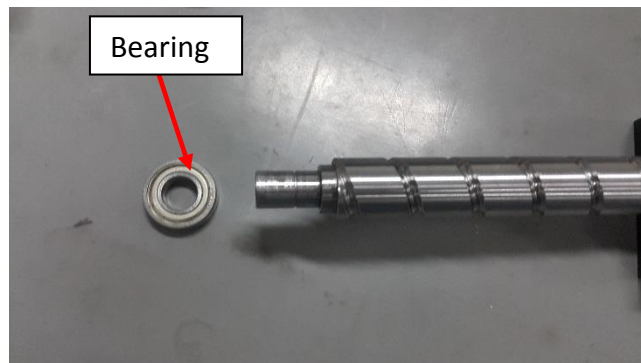


Figure 26 Bearing removal [COMPLETE]

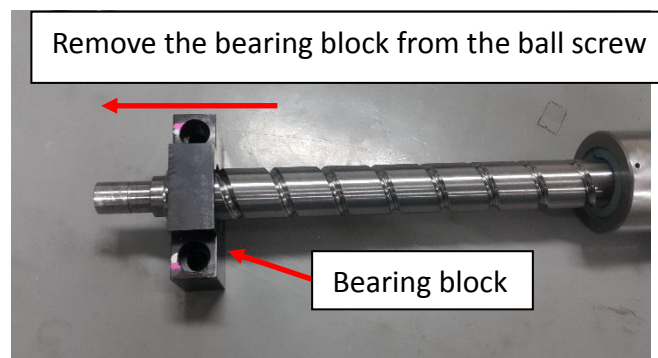


Figure 27 Bearing block

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22. By using adjustable wrench, loosen and remove the Grease Fitting, Straight, M6x13.5M (4XAXY-047) as below Figure 28.

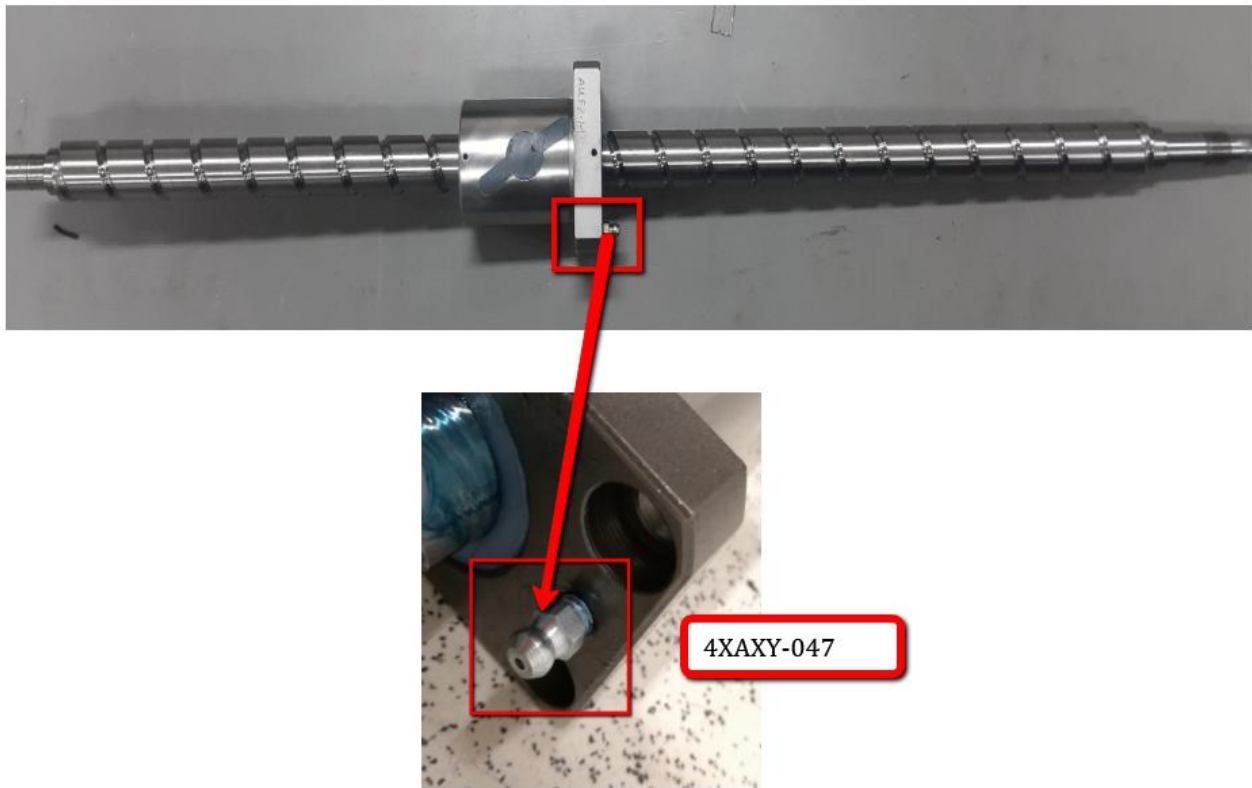


Figure 28 Grease fitting removal

X-Axis Ball Screw Replacement

23. Prepare new ball screw (By refer to below table) as below Figure 29.

Model	Part number	Part Description
V810 S1/S2	4XASPP-A022	X axis Ball Screw Kit
V810 XXL S1/S2	4XASPP-A029	XXL X axis Ball Screw Kit
V810 S2EX	4XASPP-A036	S2EX X axis Ball Screw Kit

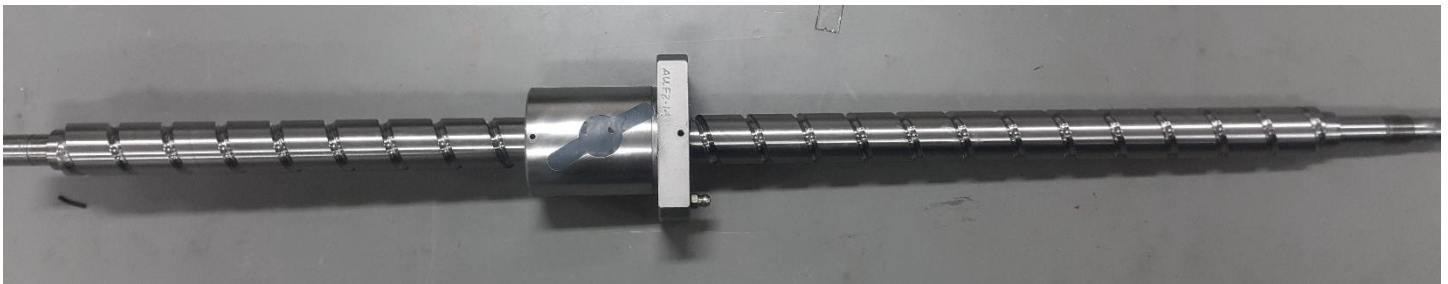


Figure 29 [4XAXY-039] X-Axis Ball Screw

24. By using adjustable wrench, assemble the Grease Fitting, Straight, M6x13.5M (4XAXY-047) as below Figure 30.

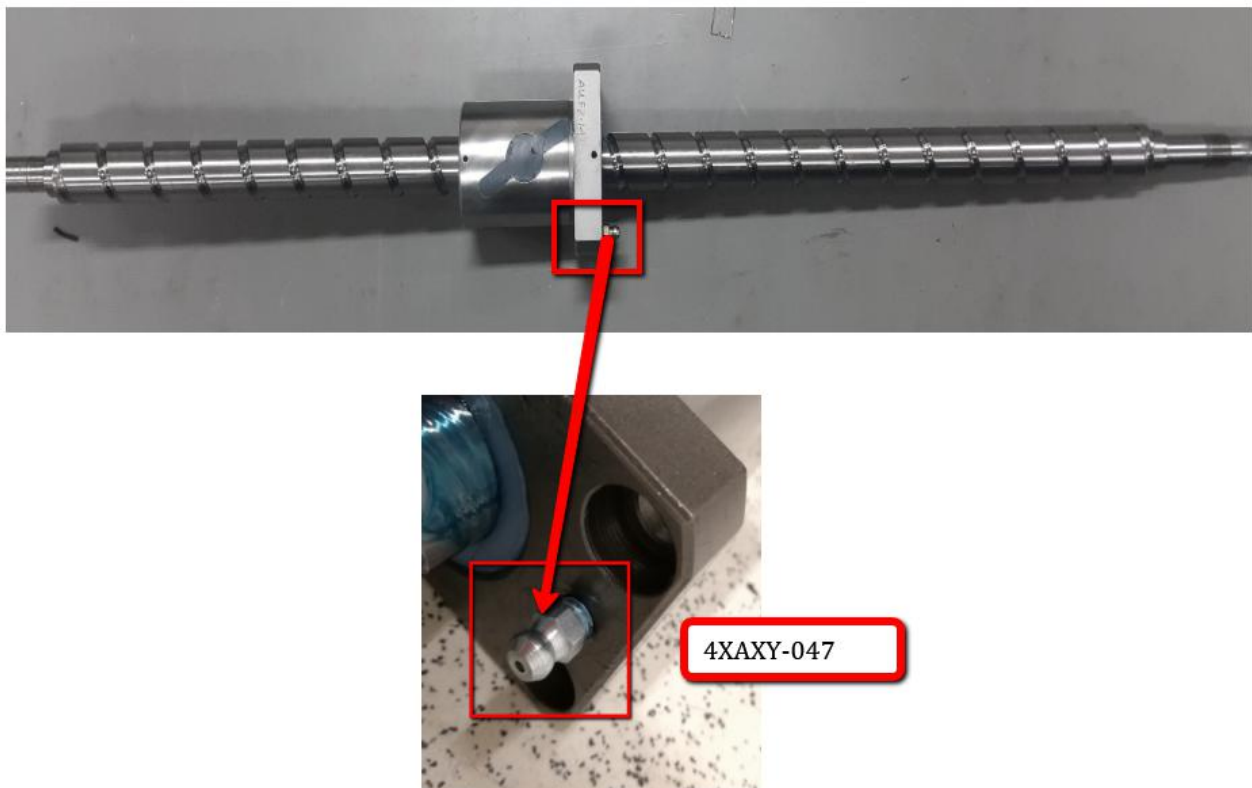


Figure 30 Grease Fitting Assembly

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25. Insert the bearing block into the new ball screw as shown in the Figure 31.

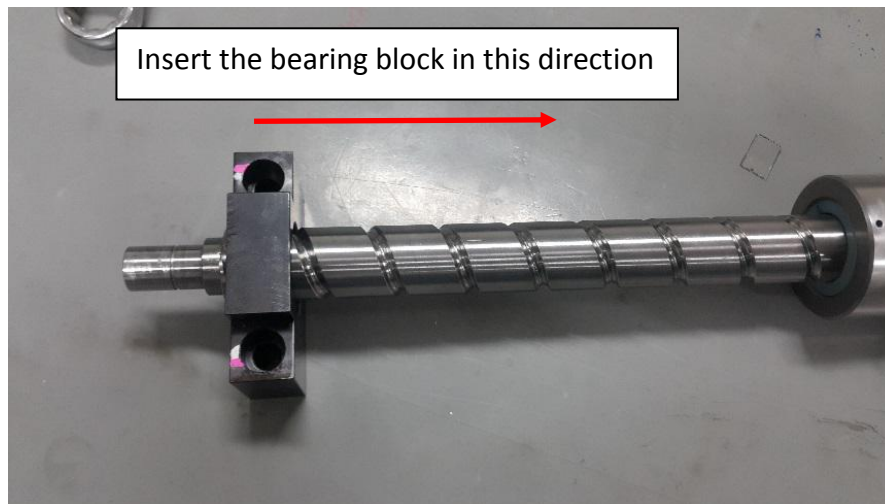


Figure 31 Bearing block

26. Insert the bearing to the ball screw as shown in the Figure 32.

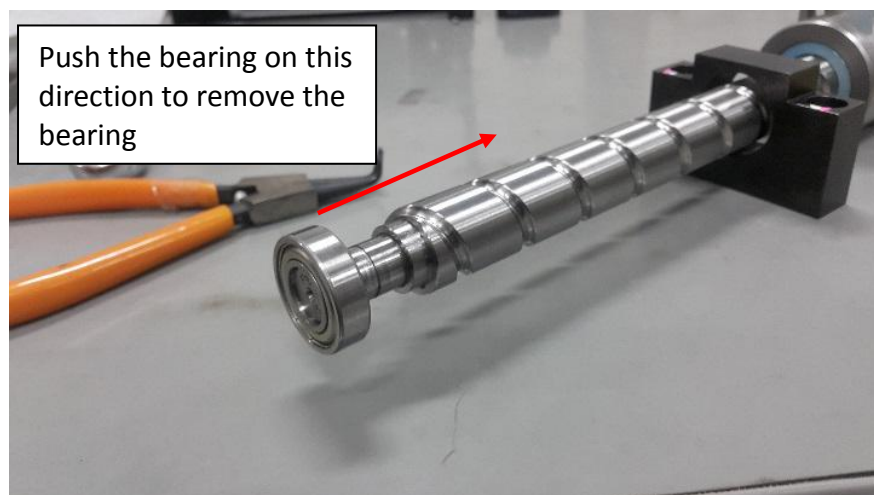


Figure 32 Bearing

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27. Using the C-Clip Plier, push the bearing until the end of the ball screw as shown in Figure 33.

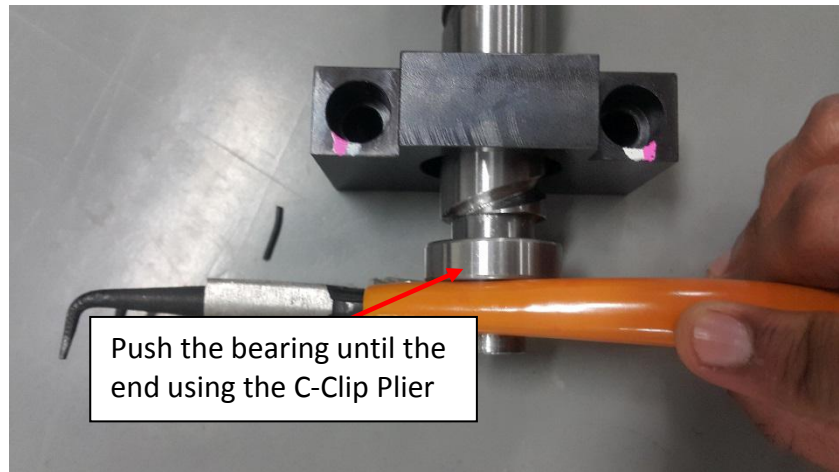


Figure 33 Bearing installation

28. Install the C-Clip as shown in the Figure 34 to Figure 36.

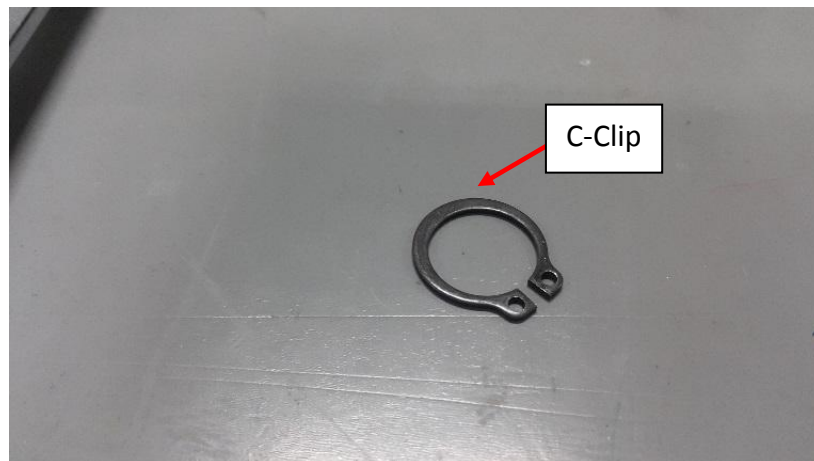


Figure 34 C-Clip

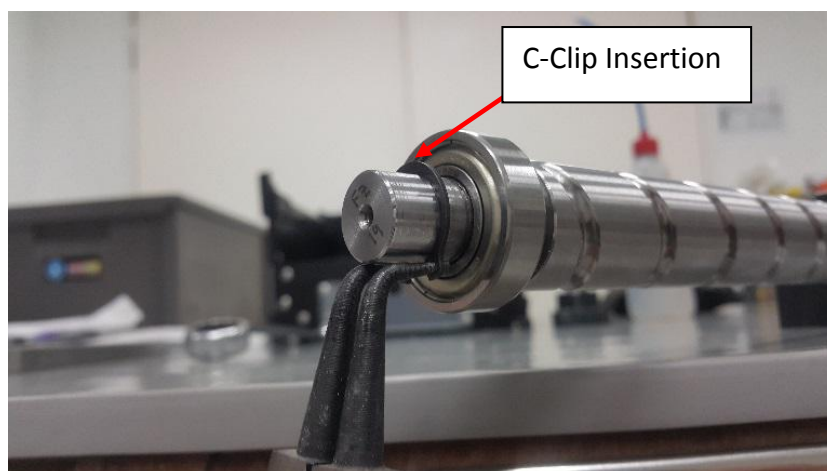


Figure 35 C-Clip Insertion

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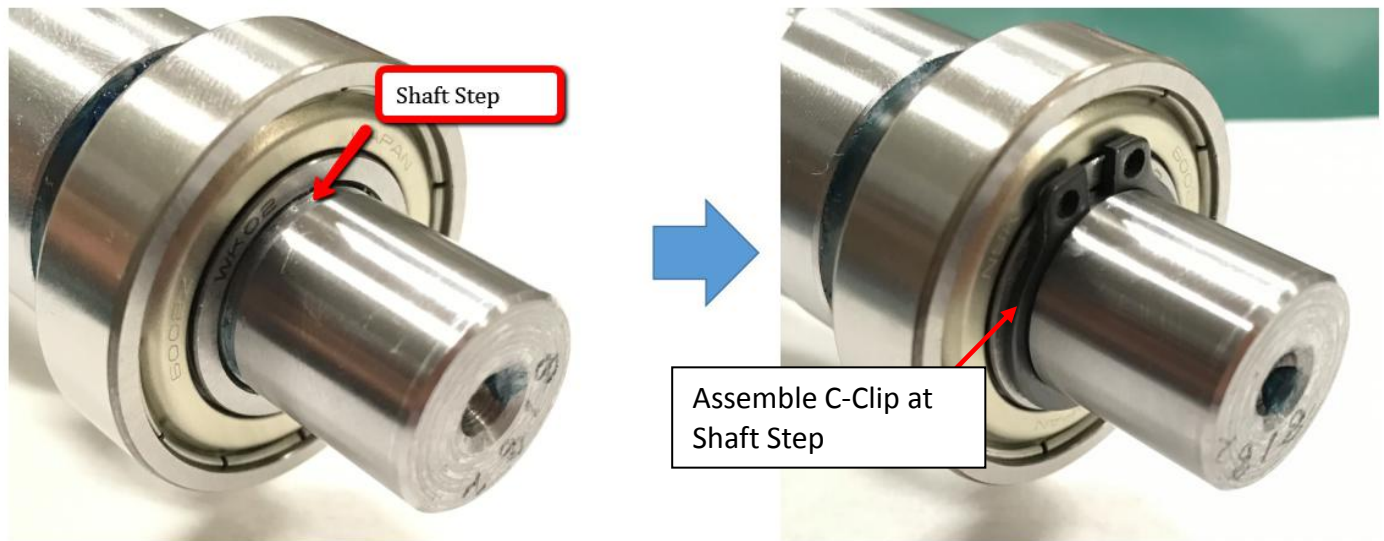


Figure 36 C-Clip Insert into Shaft Step

Remarks : C-Clip should be slot firmly into the “Shaft step” as below Figure 36

29. Position the end support bearing at front end of ball screw as Figure 37.

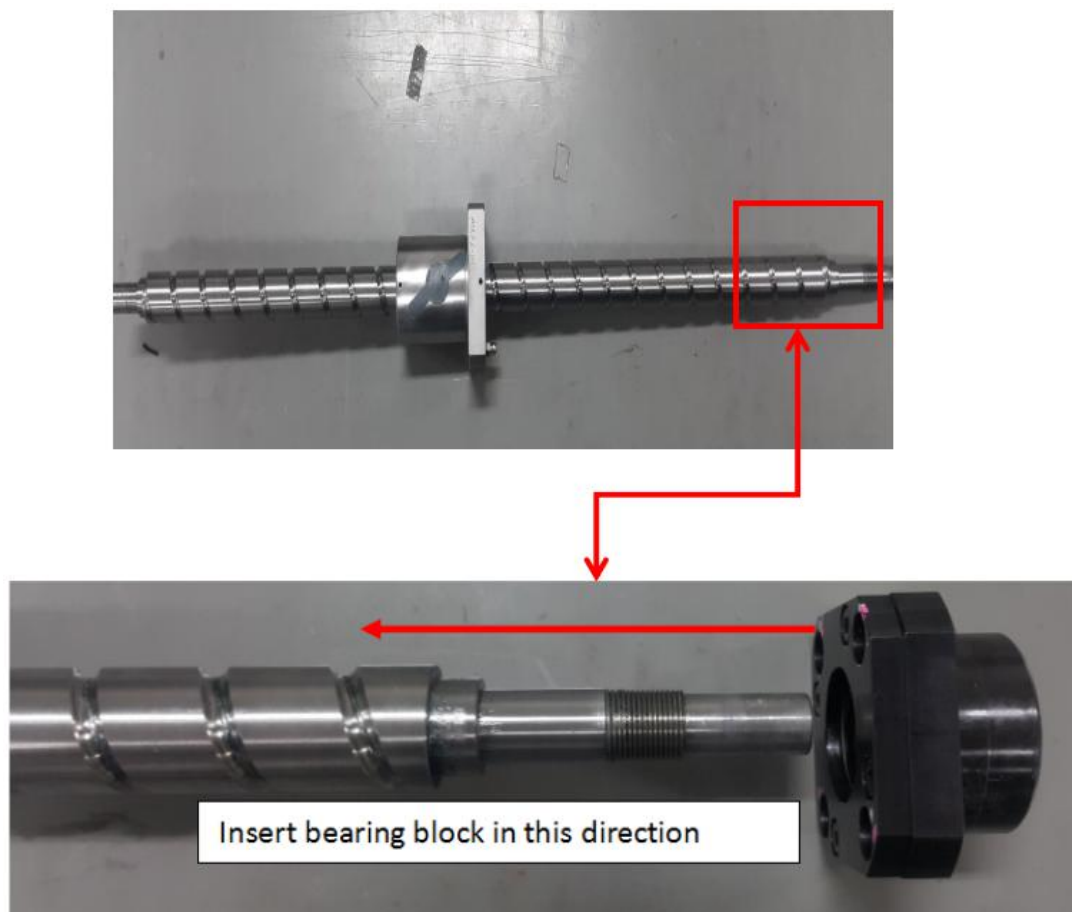


Figure 37 End Bearing Block Assembly

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30. Gently Insert the end bearing as Figure 38 and push till end of ball screw.

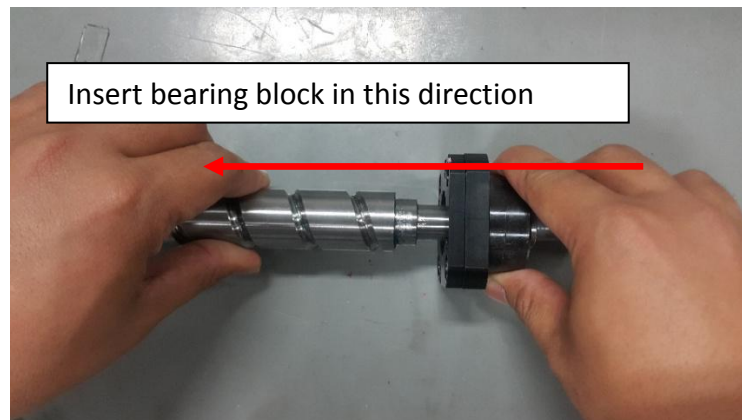


Figure 38 End bearing block

31. Gently slot in silver spacer as below Figure 39.

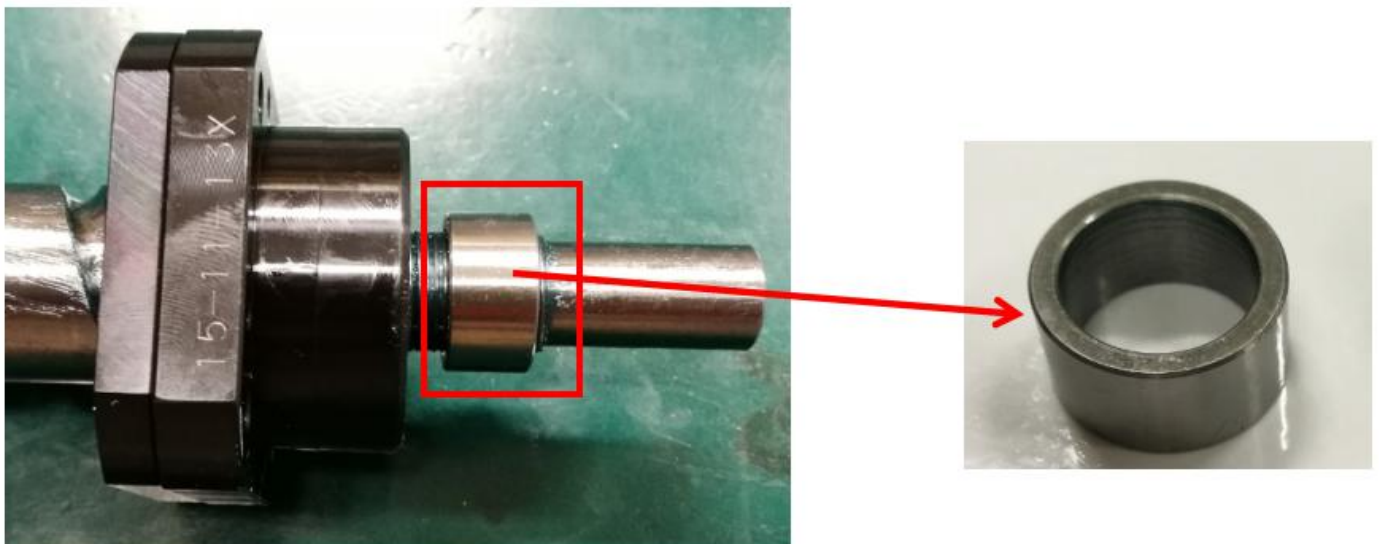


Figure 39 Silver spacer assembly

32. Prepare End Bearing Support Nut, Golden Nut, and M2 Set Screw which previously dismantled. May refer to below Figure 40.

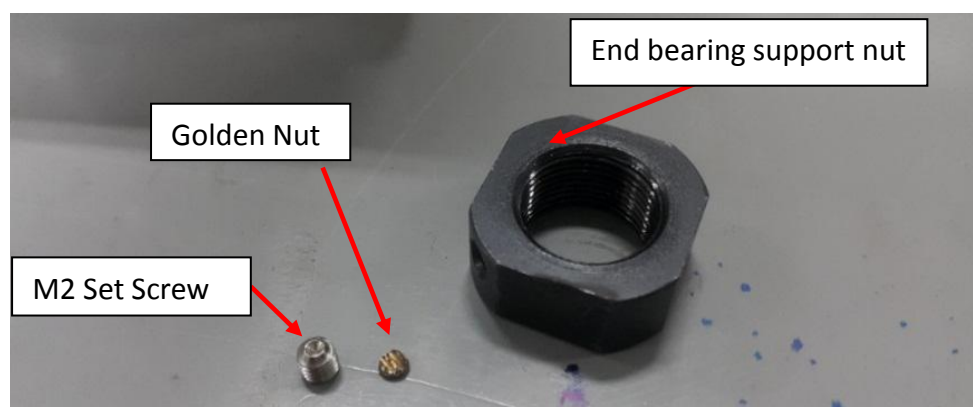


Figure 40 End Bearing Support Nut, Golden Nut, M2 Set Screw

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33. Insert the End Bearing Support Nut (refer Figure 40) to the ball screw end as shown the Figure 41.

Remarks : Support nut stack should face left direction.

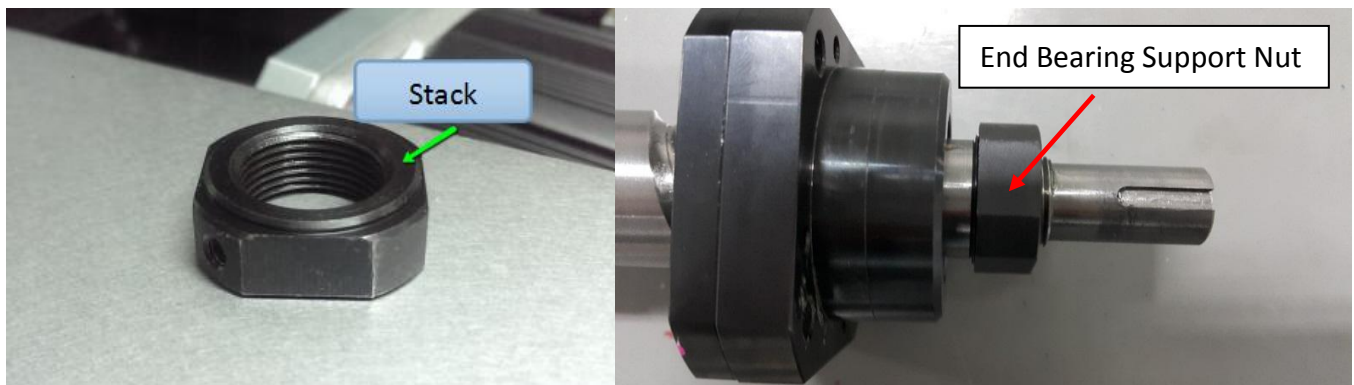


Figure 41 End Bearing Support Nut

34. Tighten the End Bearing Support Nut by using the 4XACM-A102, Ball Screw Tightening Jig with the spanner size 22 as Figure 42. Rotate the spanner in clockwise direction to tighten the End Bearing Support Nut as shown in the Figure 43.

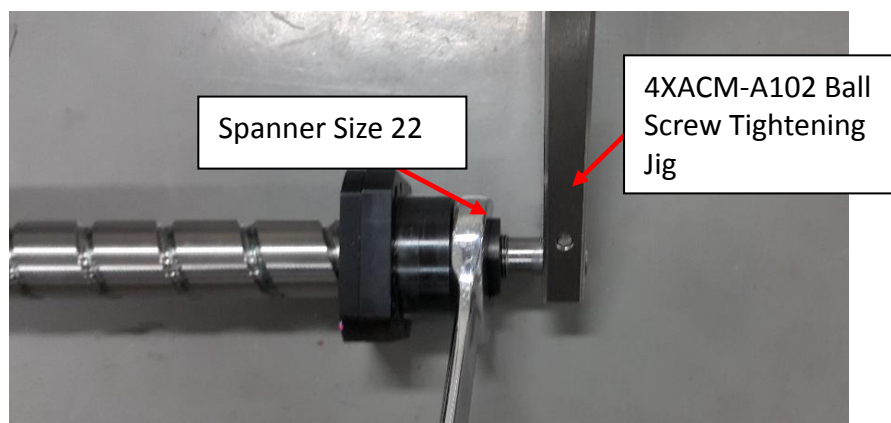


Figure 42 End bearing support nut assembly

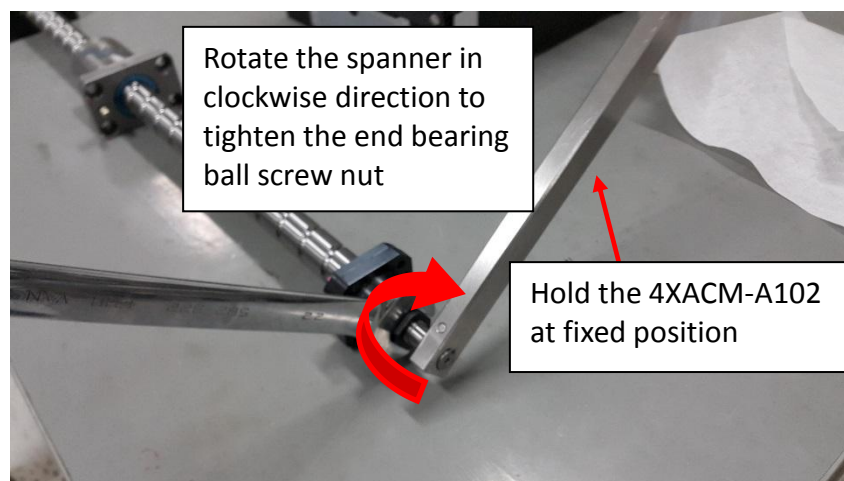


Figure 43 End bearing support nut tightening

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35. Insert the golden nut into the End Bearing Ball Screw Nut as Figure 44. Next, use M2 Allen key to tighten M2 set screw as Figure 45.

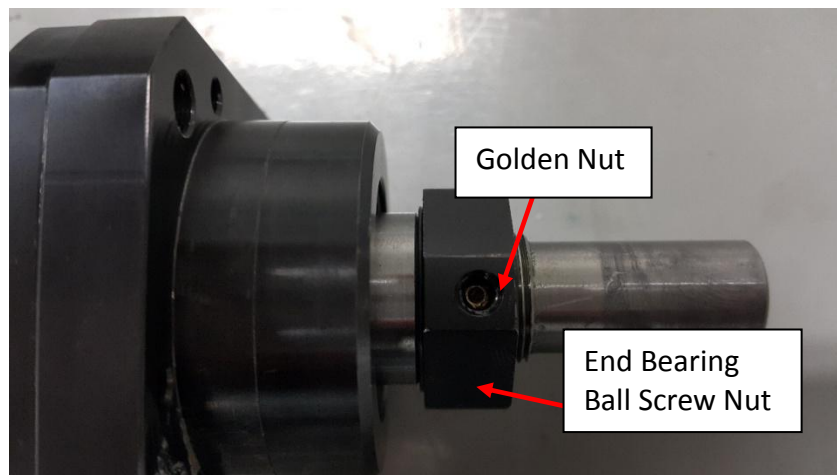


Figure 44 Golden Nut

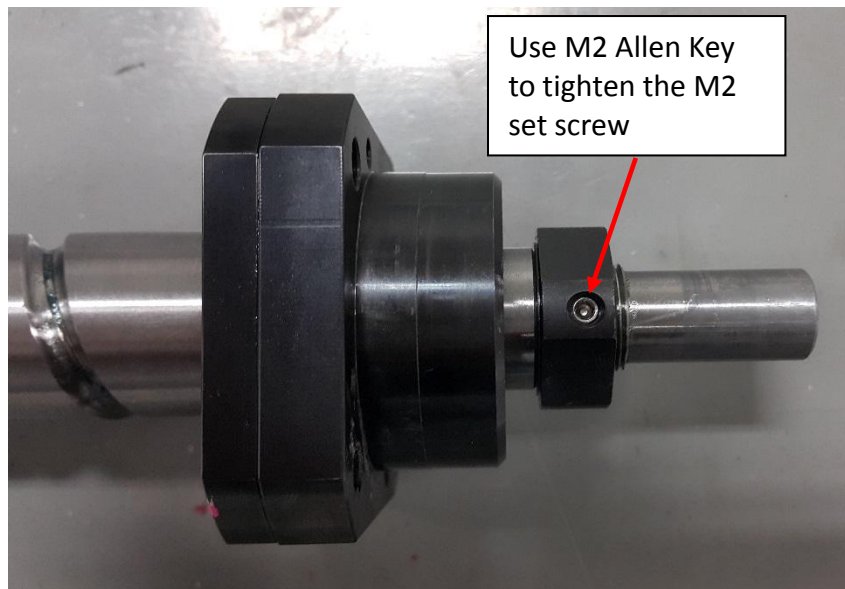


Figure 45 M2 Set Screw

36. May refer to below Figure 46 for final assembly of bearing blocks at X axis ball screw.

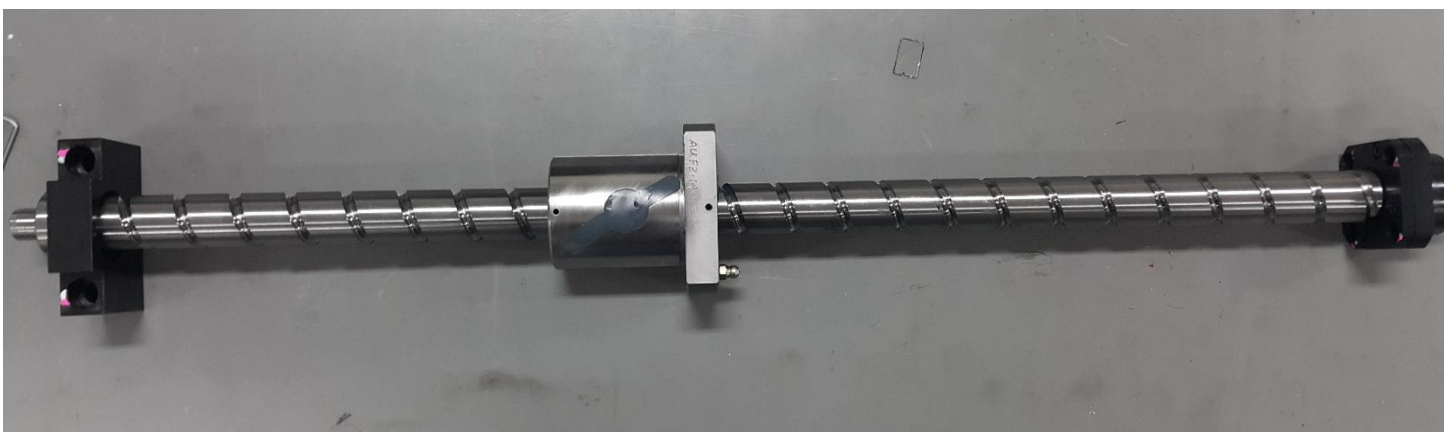


Figure 46 X-Axis Ball Screw

37. Assemble X-Axis Ball Screw at X axis stage.

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38. Install the screw at “A” & “B” of the Bearing Support with M6 Allen Key as shown in Figure 47 and do not tighten those screws.

39. Install the screw at “G” to “J” of the Fixed Support Bearing with M5 Allen Key as shown in Figure 47 and do not tighten those screws.

40. Install the screw at “C” to “F” of the Fixed Support Bearing with M6 Allen Key as shown in Figure 47 and do not tighten those screws.

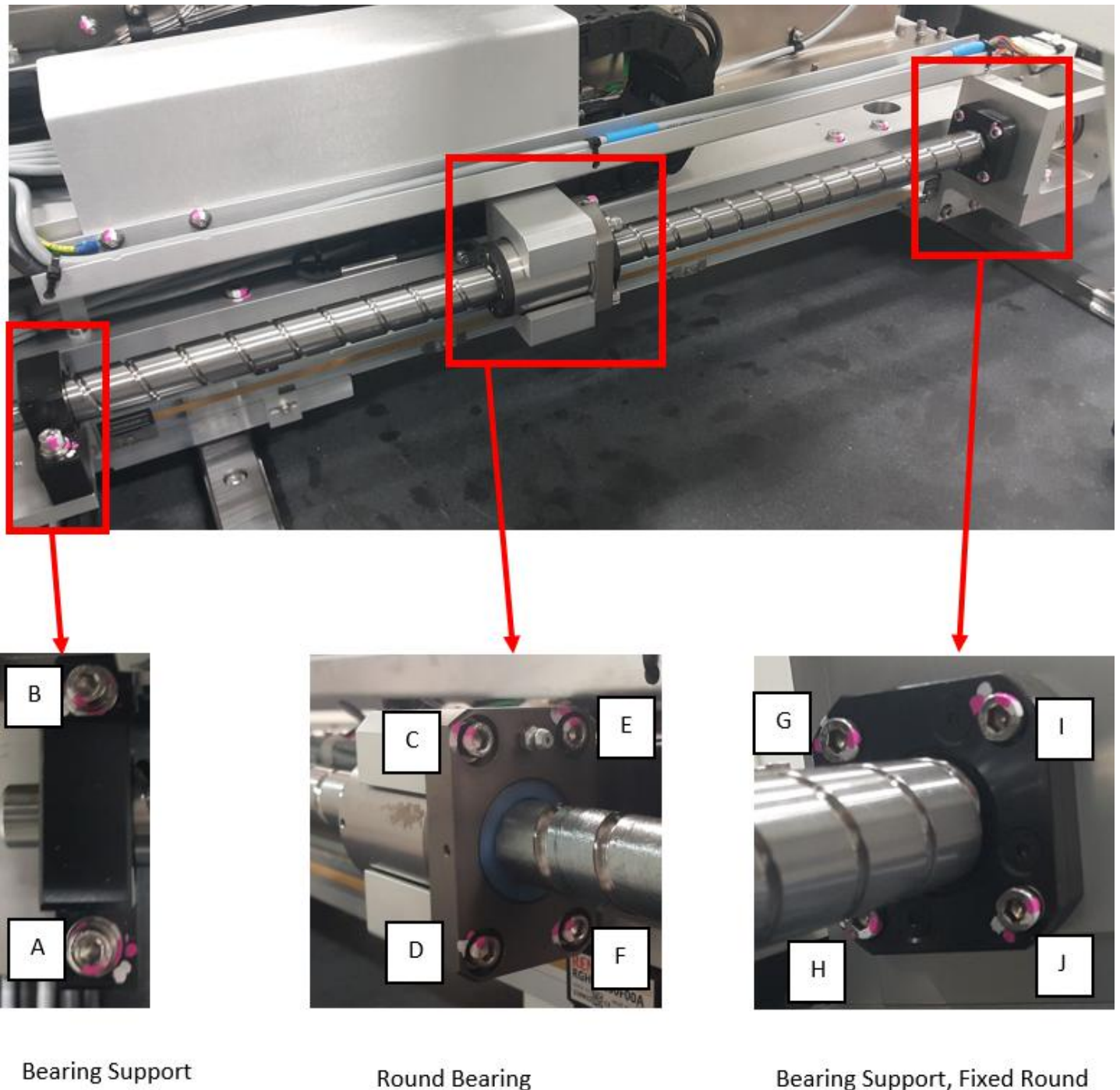


Figure 47 X-Axis Ball screw

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41. Push X-Axis stage from minimum to maximum hard stop(Left to Right)
42. Pull X-Axis stage from maximum hard stop to minimum hard stop(Right to Left)
43. Repeat Step 41~42 for 3 times.

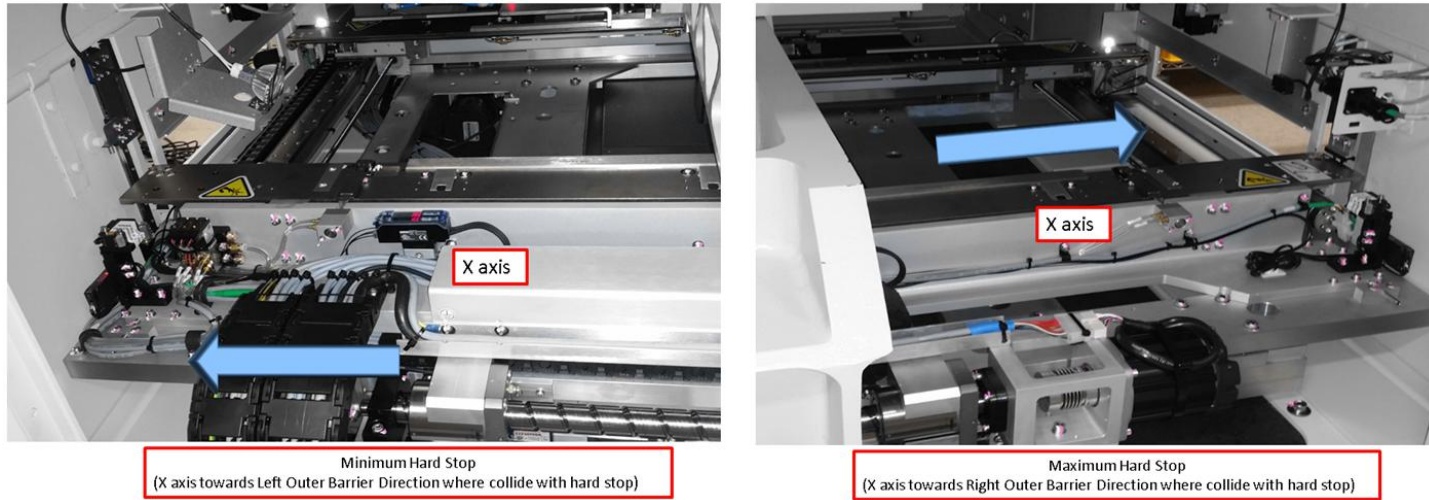


Figure 48 Minimum & Maximum Hard Stop (V810 Std)

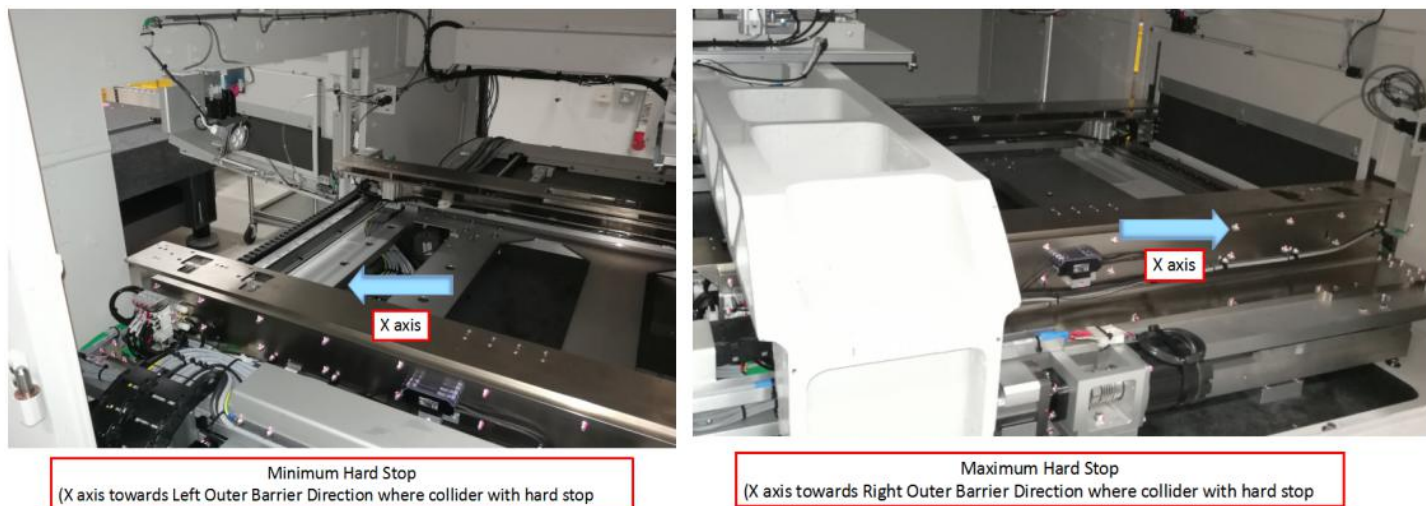


Figure 49 Minimum & Maximum Hard Stop (V810 XXL)

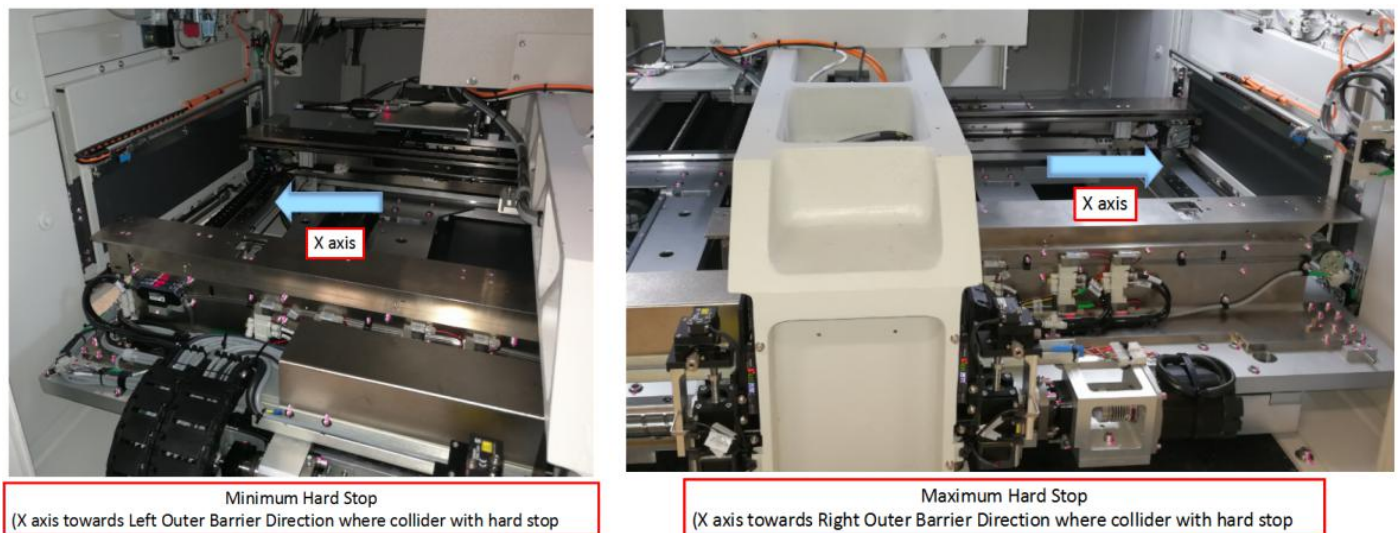
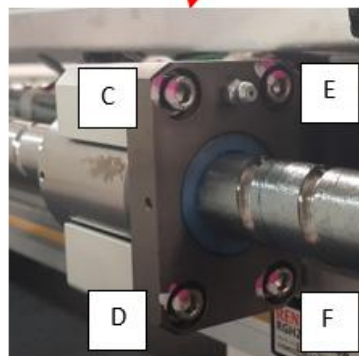
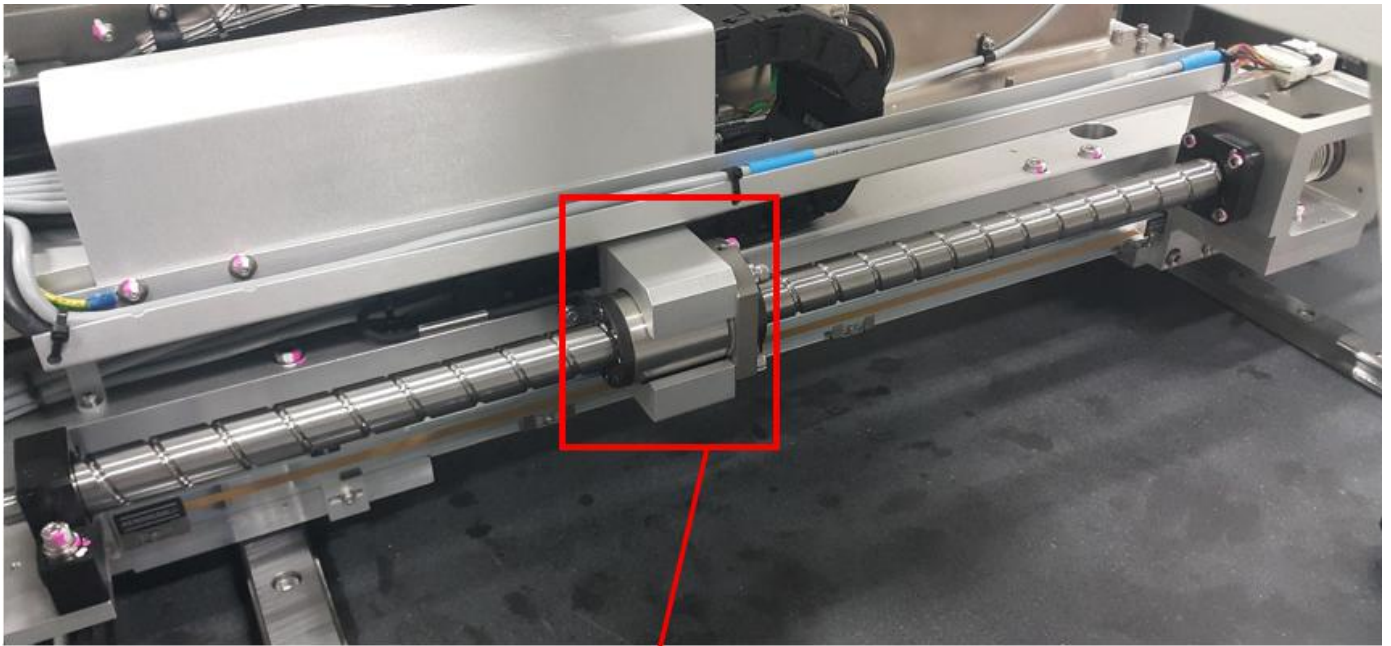


Figure 50 Minimum & Maximum Hard Stop (V810 S2EX)

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44. Push the X axis from Home position to the middle of the stage and tighten the 4 screws for the Bearing support, Fixed round as shown in Figure 51.

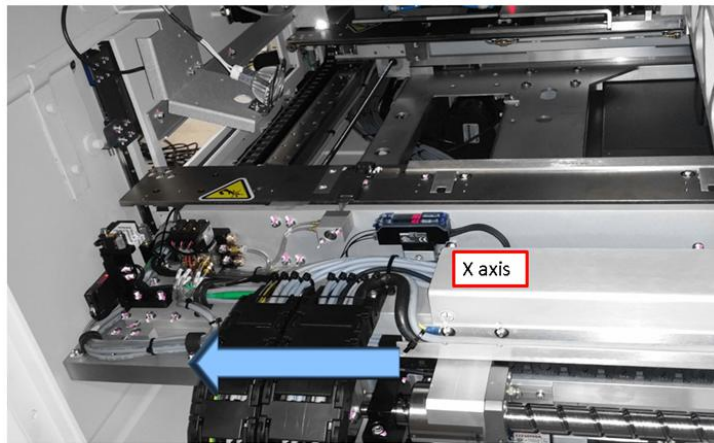


Round Bearing

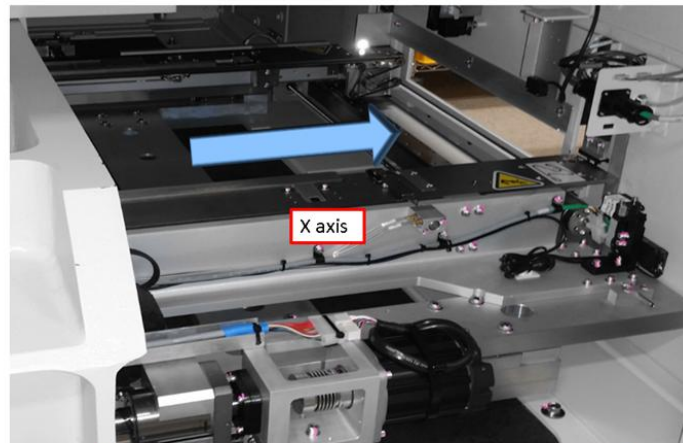
Figure 51 Middle Bearing Support

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45. Push X-Axis stage from minimum to maximum hard stop(Left to Right)
46. Pull X-Axis stage from maximum hard stop to minimum hard stop(Right to Left)
47. Repeat Step 45~46 for 3 times.



Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collide with hard stop)

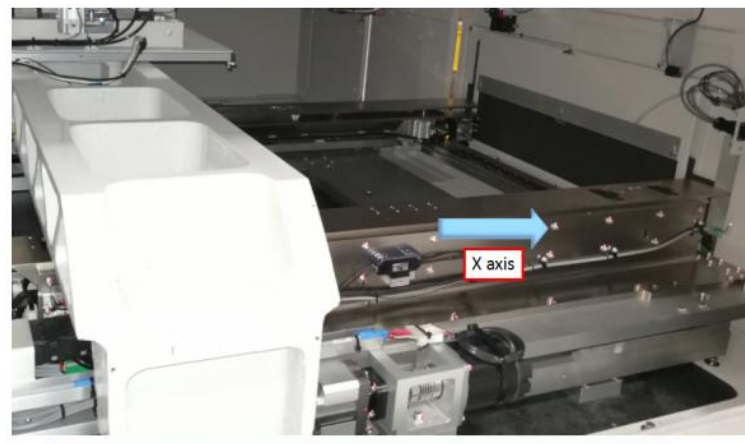


Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collide with hard stop)

Figure 52 Minimum & Maximum Hard Stop (V810 Std)

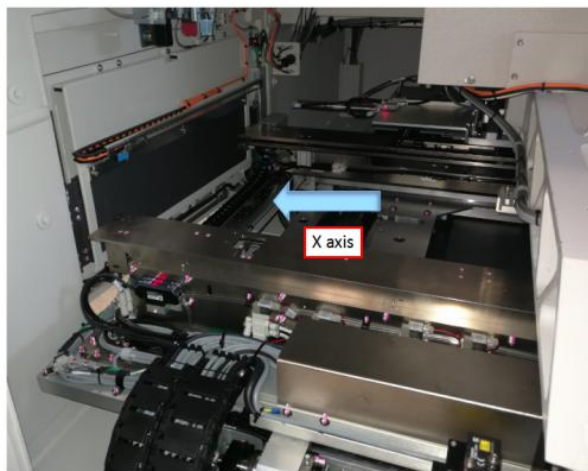


Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collider with hard stop)

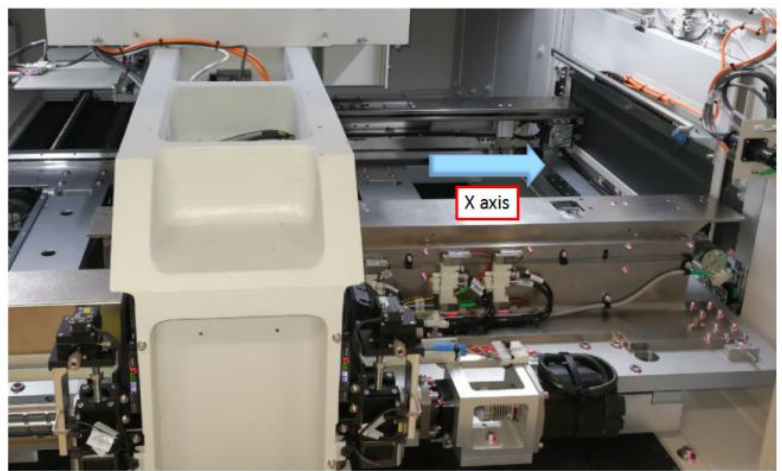


Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collider with hard stop)

Figure 53 Minimum & Maximum Hard Stop (V810 XXL)



Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collider with hard stop)

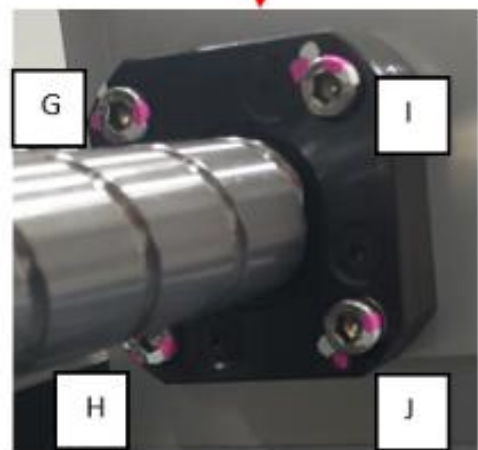
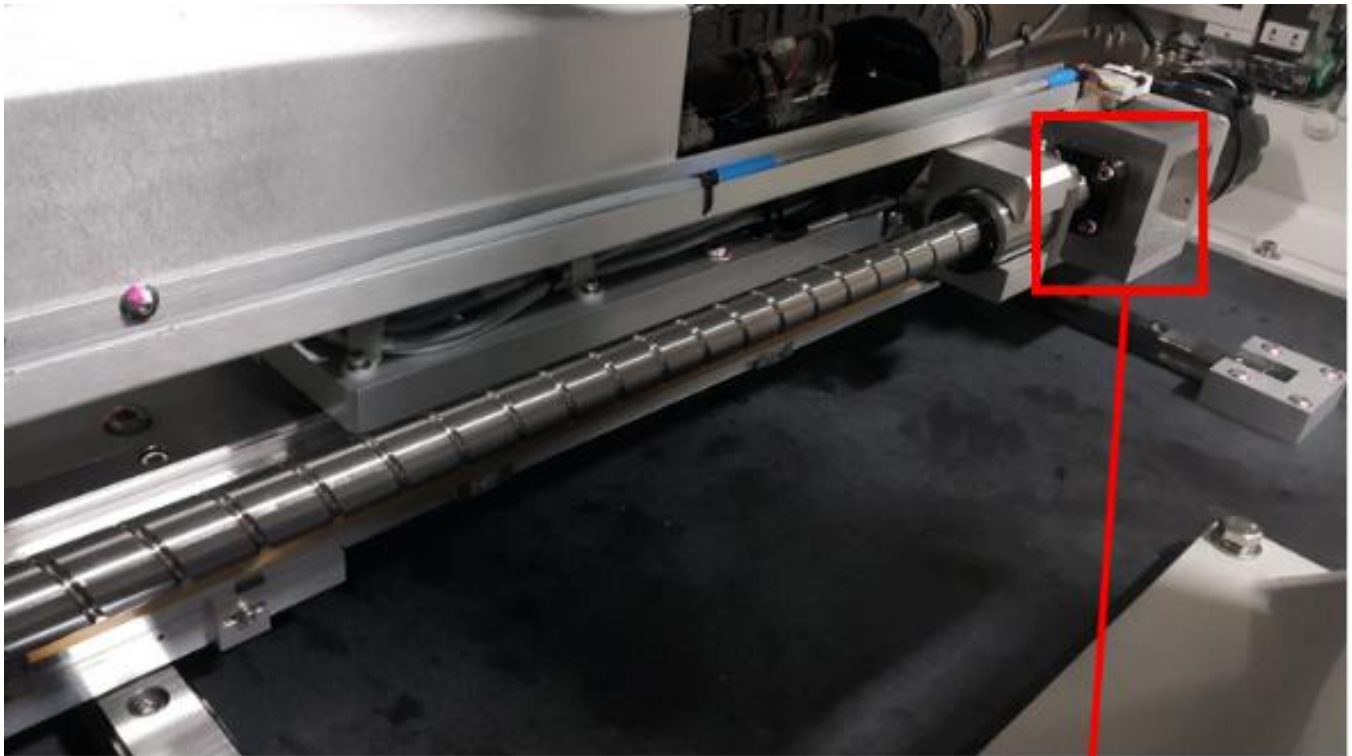


Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collider with hard stop)

Figure 54 Minimum & Maximum Hard Stop (V810 S2EX)

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48. Push the X axis stage to the maximum hard stop (Right) and tighten 4 x screws of Bearing support, Fixed round as shown in Figure 55.



Bearing Support, Fixed Round

Figure 55 End Support Bearing

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49. Push X-Axis stage from minimum to maximum hard stop(Left to Right)
50. Pull X-Axis stage from maximum hard stop to minimum hard stop(Right to Left)
51. Repeat Step 49~50 for 3 times.

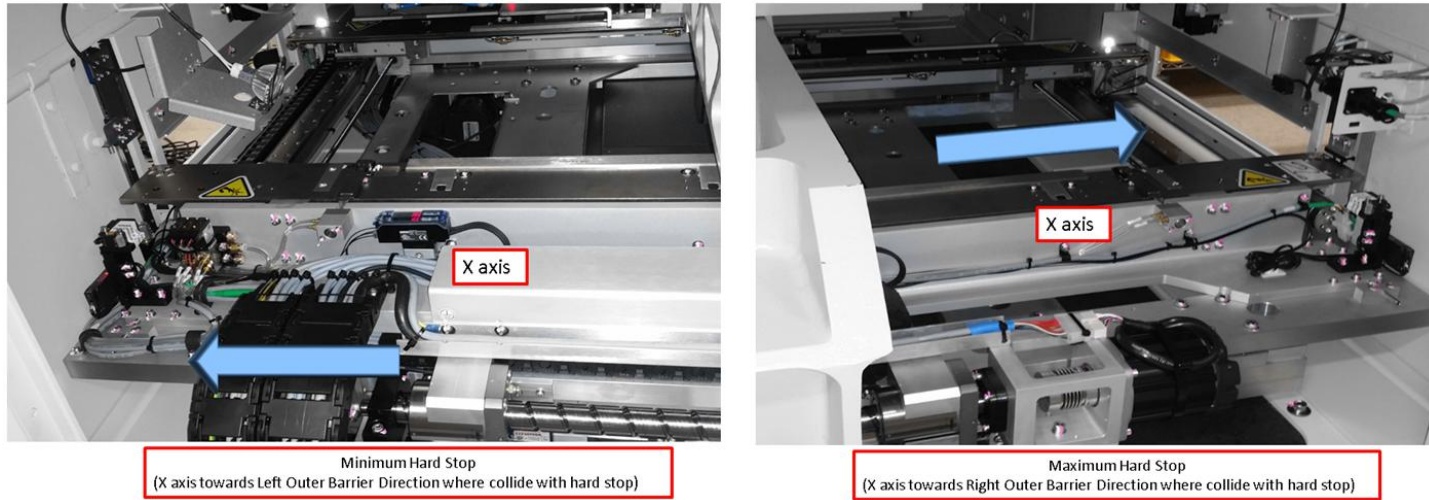


Figure 56 Minimum & Maximum Hard Stop (V810 Std)

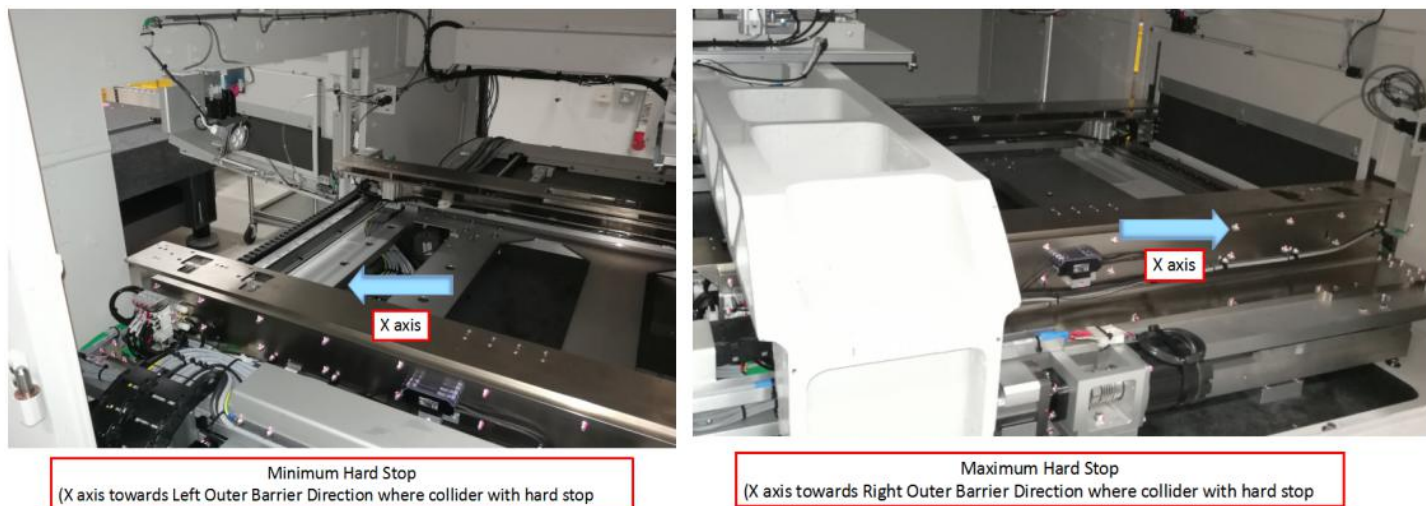


Figure 57 Minimum & Maximum Hard Stop (V810 XXL)

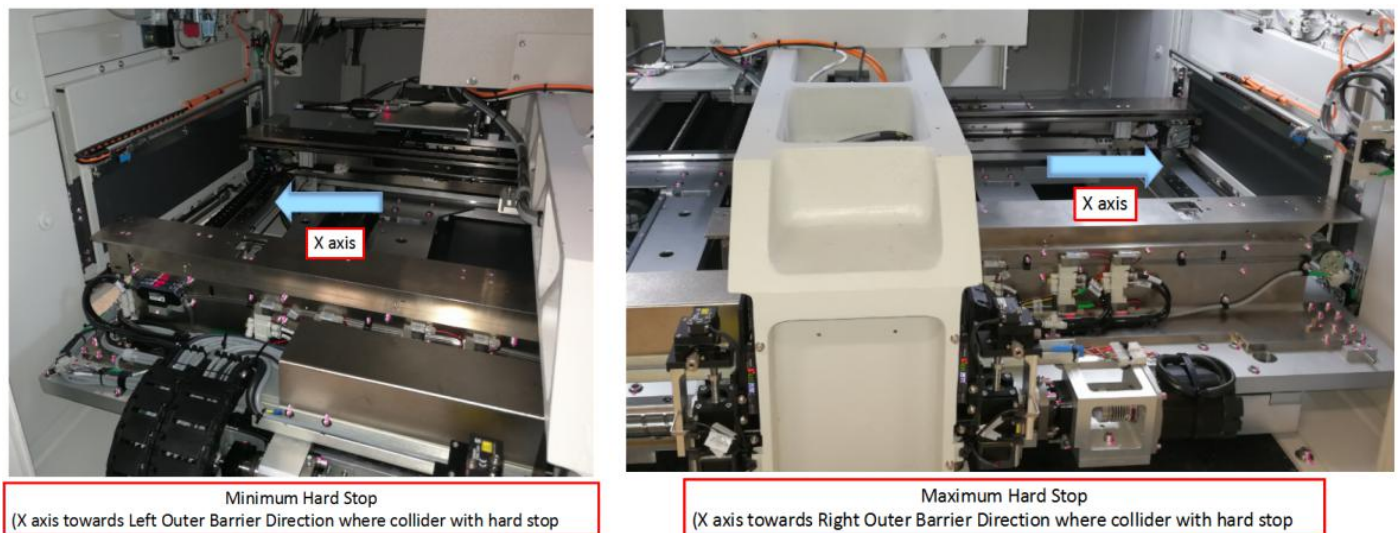


Figure 58 Minimum & Maximum Hard Stop (V810 S2EX)

	Work Instruction	Doc. Revision	03
	AXI V810 X-Axis Ball Screw Removal & Replacement Procedure	Effective Date	2 Oct 2016

52. Push the X-Axis Stage to the minimum hard stop (Left) as shown in the Figure 59 and tighten 2x screw of Bearing Support.



Bearing Support

Figure 59 End support bearing (left side)

	Work Instruction	Doc. Revision	03
	AXI V810 X-Axis Ball Screw Removal & Replacement Procedure	Effective Date	2 Oct 2016

53. Assemble & tighten up (2x Screws) coupling and X axis motor (4 x Screws) as shown in the Figure 60 and Figure 61.

Special Remarks : 2 x Coupling screws must be visible.

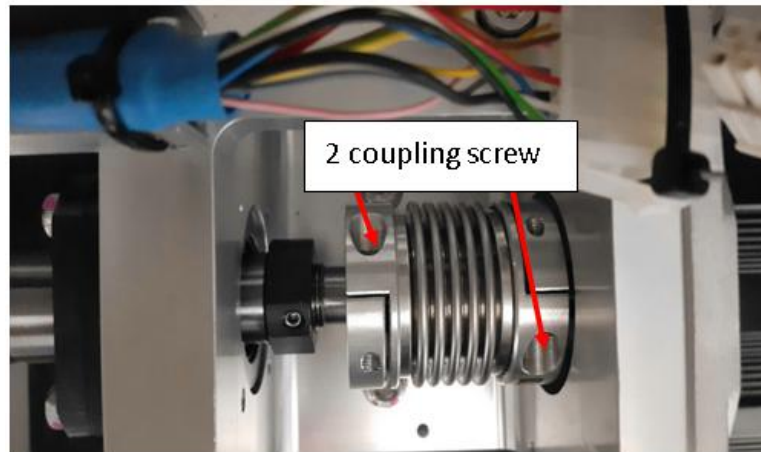


Figure 60 Coupling Screws (MUST BE VISIBLE)

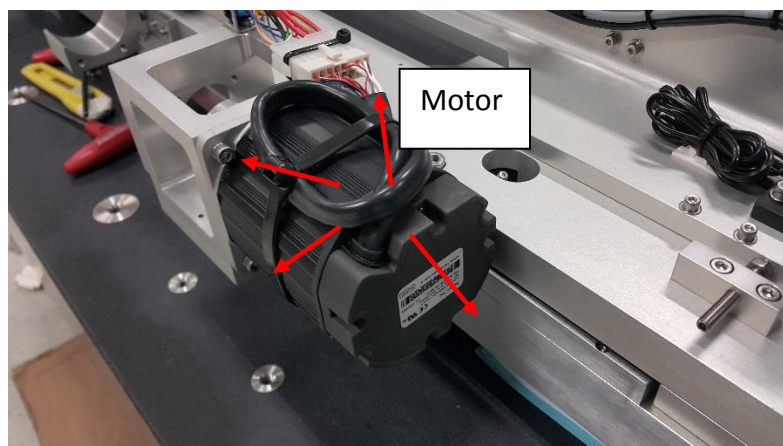


Figure 61 Motor Assembly

54. Go to the Services->Subsystem Status & Control -> Panel Positioner and Click Initialize

55. Run 5 Loop of Long Scan at Panel Positioner.

56. Run CD & A.

57. If Long Scan Or CD & A fail, may refer “ AXI V810 X-Axis Ball Screw Alignment Procedure” to perform alignment.